# NICHOLAS BALASUS

Postdoctoral Scholar, Carbon Mapper

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## EDUCATION

Ph.D., Environmental Science and Engineering, Harvard University 2021–2025

B.S., Chemical Engineering, University of Maryland, Baltimore County 2017–2021

## RESEARCH EXPERIENCE

Postdoctoral Scholar, Carbon Mapper (Advisor: Daniel Cusworth) 2025–present

Graduate Student Assistant, Harvard University (Advisor: Daniel Jacob) 2021–2025

## FIRST-AUTHOR PUBLICATIONS

1. **N. Balasus** and D.H. Cusworth (in review), Satellite observations of nitrous oxide emissions from adipic acid production, *arXiv*, doi:10.48550/arXiv.2610.07542.
2. **N. Balasus**, D.H. Cusworth, J. Kim, D.J. Varon, C.E. Miller, and R.M. Duren (2026), Mapping ammonia emission plumes using shortwave infrared imaging spectroscopy, *Proc. Natl. Acad. Sci.*, doi:10.1073/pnas.2605694123.
3. **N. Balasus**, D.J. Jacob, A.A. Bloom, J.D. East, L.A. Estrada, S.E. Hancock, M. He, T.A. Mooring, A.J. Turner, and J.R. Worden (2026), 2019–2024 trends in African livestock and wetland emissions as contributors to the global methane rise, *Atmos. Chem. Phys.*, doi:10.5194/acp-26-4601-2026.
4. **N. Balasus**, D.J. Jacob, G. Maxemin, C. Jenks, H. Nesser, J.D. Maasakkers, D.H. Cusworth, T.R. Scarpelli, D.J. Varon, and X. Wang (2025), Satellite monitoring of annual US landfill methane emissions and trends, *Environ. Res. Lett.*, doi:10.1088/1748-9326/ada2b1.
5. **N. Balasus**, D.J. Jacob, A. Lorente, J.D. Maasakkers, R.J. Parker, H. Boesch, Z. Chen, M.M. Kelp, H. Nesser, and D.J. Varon (2023), A blended TROPOMI+GOSAT satellite data product for atmospheric methane using machine learning to correct retrieval biases, *Atmos. Meas. Tech.*, doi:10.5194/amt-16-3787-2023.
6. **N. Balasus**, M.A. Battaglia Jr., K. Ball, V. Caicedo, R. Delgado, A.G. Carlton, and C.J. Hennigan (2021), Urban aerosol chemistry at a land-water transition site during summer – Part 1: Impact of agricultural and industrial ammonia emissions, *Atmos. Chem. Phys.,* doi:10.5194/acp-21-13051-2021.

## CO-AUTHOR PUBLICATIONS

1. S.E. Hancock, D.J. Jacob, **N. Balasus**, C.A. Garzon, C.C. Junior, M.R. Johnson, N. Calderón-Cangrejo, S. Winton, M. He, L. Potyk, L.A. Estrada, R. Jimenez, A. Ardila, L. Morales-Rincon, and J.L. France (in review), Explaining the rapid 2019–2024 increase in methane emissions from Colombia seen in satellite observations: a high-resolution inverse analysis.
2. K.D. Gauld, D.J. Varon, **N. Balasus**, and D.H. Cusworth (in review), Quantifying AI data center nitrogen oxide ($NO_x$) emissions from space.
3. D. Zhang, D.J. Jacob, E.W. Lundgren, M.P. Sulprizio, **N. Balasus**, L.A. Estrada, E. Reidy, and D.J. Varon (in review), Efficient computation of global high-resolution Jacobian matrices for analytical inversions of satellite observations of greenhouse gases.
4. M.L. Loman, L.T. Murray, P.D. Ivatt, R. Commane, A. Hallward-Driemeier, R. Dhall, E.M. Leibensperger, **N. Balasus**, L. Estrada, M.P. Sulprizio, M.R. Sargent, and F. Vogel (in review), Assessing the ability of stationary *in situ* ground-based observations to constrain local methane emissions in space and time in New York State.
5. S. Lama, J.D. Maasakkers, X. Zhang, M. Girard, S. Dutt, S. Nandgaonkar, D.J. Varon, M.P. Sulprizio, L.A. Estrada, **N. Balasus**, R.J. Parker, Y. Terao, and I. Aben (in review), Evaluating national, state, and urban Indian methane emissions using satellites.
6. S. Baray, F.R. Vogel, D.J. Varon, L.A. Estrada, **N. Balasus**, D.B.A. Jones, C.A. McLinden, D. Griffin, M. Watine-Guiu, R.M. Duren, D.H. Cusworth, T.A. de Jong, and J.D. Maasakkers (in review), Estimating methane emissions from surface coal mines using complementary satellite observations.

7. L.A. Estrada, D.J. Jacob, M. He, J.D. East, D.J. Varon, **N. Balasus**, S.E. Hancock, M. Sulprizio, K.W. Bowman, J.R. Worden, E. Reidy, and B.R.K. Runkle (2026), Quantifying national, state, and oil/gas field methane emissions and trends in the U.S., *Atmos. Chem. Phys.*, doi:10.5194/acp-26-10629-2026.
8. S.E. Hancock, D.J. Jacob, R. Jimenez, A. Ardila, L. Morales-Rincon, N. Rojas, L.A. Estrada, **N. Balasus**, J.D. East, M.P. Sulprizio, X. Wang, J.L. France, L. Potyk, E. Penn, Z. Chen, D.J. Varon, C. Frankenberg, M. Baranski, A. Calcan, and R.J. Parker (2026), Applying satellite observations to improve bottom-up national emission inventories for methane: application to Colombia, *Atmos. Chem. Phys.*, doi:10.5194/acp-26-10455-2026.
9. Z. Chen, D.J. Jacob, H. Lin, **N. Balasus**, S.E. Hancock, L.A. Estrada, J.D. East, Y. Zhang, X. Wang, M. He, M. Liu, and D.J. Varon (2026), Tropical wetland methane emissions and trends (2004–2023) inferred from Landsat-based inundated vegetation data, *Environ. Sci. Technol.*, doi:10.1021/acs.est.6c05412.
10. M. He, D.J. Jacob, L.A. Estrada, D.J. Varon, M. Sulprizio, **N. Balasus**, J.D. East, E. Penn, D.C. Pendergrass, Z. Chen, T.A. Mooring, J.D. Maasakkers, P.G. Brodrick, C. Frankenberg, K.W. Bowman, and L. Bruhwiler (2026), Attributing 2019–2024 methane growth using TROPOMI satellite observations, *Sci. Adv.*, doi:10.1126/sciadv.adz9007.
11. X. Wang, D.J. Jacob, H. Nesser, **N. Balasus**, L.A. Estrada, M.P. Sulprizio, D.H. Cusworth, T.R. Scarpelli, Z. Chen, J.D. East, and D.J. Varon (2026), Quantifying urban and landfill methane emissions in the United States using TROPOMI satellite data, *Sci. Adv.*, doi:10.1126/sciadv.adz9308.
12. D. Varon, D.J. Jacob, L.A. Estrada, **N. Balasus**, J.D. East, D.C. Pendergrass, Z. Chen, M. Sulprizio, M. Omara, R. Gautam, Z.R. Barkley, F.J. Cardoso-Saldaña, E.K. Reidy, H. Kamdar, E.D. Sherwin, S.C. Biraud, D. Jervis, S. Pandey, J.R. Worden, K.W. Bowman, J.D. Maasakkers, and R.L. Kleinberg (2025), Seasonality and declining intensity of methane emissions from the Permian and nearby US oil and gas basins, *Environ. Sci. Technol.*, doi:10.1021/acs.est.5c08745.
13. J.D. East, D.J. Jacob, D. Jervis, **N. Balasus**, L.A. Estrada, S.E. Hancock, M.P. Sulprizio, J. Thomas, X. Wang, Z. Chen, D.J. Varon, and J.R. Worden (2025), Worldwide inference of national methane emissions by inversion of satellite observations and UNFCCC prior estimates, *Nat. Commun.*, doi:10.1038/s41467-025-67122-8.
14. D.C. Pendergrass, D.J. Jacob, **N. Balasus**, L.A. Estrada, D.J. Varon, J.D. East, M. He, T.A. Mooring, E. Penn, H. Nesser, and J.R. Worden (2025), Trends and seasonality of 2019–2023 global methane emissions inferred from a localized ensemble transform Kalman filter (CHEEREIO v1.3.1) applied to TROPOMI satellite observations, *Atmos. Chem. Phys.*, doi:10.5194/acp-25-14353-2025.
15. L.A. Estrada, D.J. Varon, M. Sulprizio, H. Nesser, Z. Chen, **N. Balasus**, S.E. Hancock, M. He, J.D. East, T.A. Mooring, A.O. Alonso, J.D. Maasakkers, I. Aben, S. Baray, K.W. Bowman, J.R. Worden, F.J. Cardoso-Saldaña, E. Reidy, and D.J. Jacob (2025), Integrated Methane Inversion (IMI) 2.0: an improved research and stakeholder tool for monitoring total methane emissions with high resolution worldwide using TROPOMI satellite observations, *Geosci. Model Dev.*, doi:10.5194/gmd-18-3311-2025.
16. Z. Chen, H. Lin, **N. Balasus**, A. Hardy, J.D. East, Y. Zhang, B.R.K. Runkle, S.E. Hancock, C.A. Taylor, X. Du, B.O. Sander, and D.J. Jacob (2025), Global Rice Paddy Inventory (GRPI): a high-resolution inventory of methane emissions from rice agriculture based on Landsat satellite inundation data, *Earth's Future*, doi:10.1029/2024EF005479.
17. S.E. Hancock, D.J. Jacob, Z. Chen, H. Nesser, A. Davitt, D.J. Varon, M.P. Sulprizio, **N. Balasus**, L.A. Estrada, J.D. East, E. Penn, C.A. Randles, J. Worden, I. Aben, R.J. Parker, and J.D. Maasakkers (2025), Satellite quantification of methane emissions from South American countries: A high-resolution inversion of TROPOMI and GOSAT observations, *Atmos. Chem. Phys.*, doi:10.5194/acp-25-797-2025.
18. Y.J. Oak, D.J. Jacob, **N. Balasus**, L.H. Yang, H. Chong, J. Park, H. Lee, G.T. Lee, E.S. Ha, R.J. Park, H. Kwon, and J. Kim (2024), A bias-corrected GEMS geostationary satellite product for nitrogen dioxide using machine learning to enforce consistency with the TROPOMI satellite instrument, *Atmos. Meas. Tech.*, doi:10.5194/amt-17-5147-2024.

19. D.J. Varon, D. Jervis, S. Pandey, S.L. Gallardo, **N. Balasus**, L.H. Yang, D.J. Jacob (2024), Quantifying $NO_x$ point sources with Landsat and Sentinel-2 satellite observations of $NO_2$ plumes, *Proc. Natl. Acad. Sci.*, doi:10.1073/pnas.2317077121.
20. J.D. East, D.J. Jacob, **N. Balasus**, A.A. Bloom, L. Bruhwiler, Z. Chen, J.O. Kaplan, L.J. Mickley, T.A. Mooring, E. Penn, B. Poulter, M.P. Sulprizio, J.R. Worden, R.M. Yantosca, and Z. Zhang (2024), Interpreting the seasonality of atmospheric methane, *Geophys. Res. Lett.*, doi:10.1029/2024GL108494.
21. Z. Chen, **N. Balasus**, H. Lin, H. Nesser, and D.J. Jacob (2024), African rice cultivation linked to rising methane, *Nat. Clim. Chang.*, doi:10.1038/s41558-023-01907-x.
22. M. Watine-Guiu, D.J. Varon, I. Irakulis-Loitxate, **N. Balasus**, and D.J. Jacob (2023), Geostationary satellite observations of extreme and transient methane emissions from oil and gas infrastructure, *Proc. Natl. Acad. Sci.*, doi:10.1073/pnas.2310797120.
23. M.A. Battaglia Jr., **N. Balasus**, K. Ball, V. Caicedo, R. Delgado, A.G. Carlton, and C.J. Hennigan (2021), Urban aerosol chemistry at a land-water transition site during summer – Part 2: Aerosol pH and liquid water content, *Atmos. Chem. Phys.*, doi:10.5194/acp-21-18271-2021.

## TEACHING EXPERIENCE

**Teaching Assistant at Harvard University**

- Writing for Scientists & Engineers (Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025)
- Confronting Climate Change: A Foundation in Science, Technology, and Policy (Fall 2022)

**Teaching Assistant at University of Maryland, Baltimore County**

- Chemical Process Control and Safety (Spring 2021)
- Chemical Process Thermodynamics (Fall 2020)
- Chemical Engineering Problem Solving and Experimental Design Lab (Spring 2020)
- Chemical Engineering Analysis (Fall 2019)

## HONORS

National Defense Science and Engineering Graduate Fellowship 2022
NOAA Ernest F. Hollings Scholar 2019

## SERVICE

Reviewer for *Atmospheric Environment*, *Atmospheric Measurement Techniques*, *Earth System Science Data*, *Environmental Research Letters*, *Geophysical Research Letters*, *Nature Climate Change*, *Nature Communications*, and *Remote Sensing of Environment*.